The CMB Framework: A Hybrid Chaotic–Metaheuristic–Blockchain Architecture for Verifiable Transform-Domain Image Steganography

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Abstract. The capacity–imperceptibility–robustness trade-off remains a foundational mathematical challenge in digital image steganography. While transform-domain embedding techniques and evolutionary metaheuristics effectively optimize coefficient selection to conceal data, they inherently lack out-of-band mechanisms to detect post-transmission carrier tampering across unsecured channels. This paper proposes the Chaotic–Metaheuristic–Blockchain (CMB) framework, a multi-tiered architecture engineered for verifiable image steganography that simultaneously addresses payload confidentiality, embedding distortion minimization, and post-transmission integrity assurance.

To ensure payload confidentiality and statistical diffusion prior to embedding, we employ the continuous-time three-dimensional Lorenz chaotic system to generate a high-entropy keystream for bit-level payload scrambling across a theoretical keyspace of approximately 2^{279}. To minimize visual and statistical distortion during embedding, we operate exclusively within the high-frequency sub-bands of the Haar Integer Wavelet Transform (IWT) domain, utilizing a hybrid Genetic Algorithm (GA) and Tabu Search (TS) metaheuristic. The GA performs an expansive global search to identify optimal coefficient subsets, while the TS executes localized neighborhood refinements to actively locate exact value-matches, thereby driving the Mean Squared Error (MSE) toward theoretical minimums. Finally, to address the critical vulnerability of active carrier modification, a private Ethereum-based blockchain layer anchors the Keccak-256 cryptographic hash of the finalized stego-image, enabling deterministic, independent receiver-side tamper detection.

This paper provides the complete mathematical formulation, comprehensive algorithmic design with pseudocode, computational complexity analysis, and a rigorous experimental evaluation protocol demonstrating theoretical resilience against both statistical steganalysis and active channel tampering. The CMB framework demonstrates measurable improvements in PSNR and SSIM over comparable baselines while uniquely providing blockchain-backed tamper detection—a contribution absent from all surveyed prior art.

Keywords: Image Steganography · Chaotic Dynamical Systems · Lorenz Attractor · Genetic Algorithm · Tabu Search · Integer Wavelet Transform · Blockchain Integrity · Keccak-256 · Steganalysis · Ethereum.

1 Introduction

The proliferation of ubiquitous, unsecured digital communication networks has made the development of robust information assurance mechanisms an urgent priority. Standard cryptographic

protocols—including AES, RSA, and ECC—ensure data confidentiality by transforming structured plaintext into entropic ciphertext. While mathematically secure against brute-force decryption, this transformation is inherently statistically anomalous. In adversarial environments governed by Kerckhoffs' principle, this visible randomization acts as a beacon, alerting passive wardens to the existence of sensitive communication and exposing the transmission to targeted traffic analysis, metadata profiling, and censorship-based suppression. The discipline of steganography complements classical cryptography by concealing the very existence of the communication within innocuous carrier media such as digital images, audio streams, or video sequences.

The term steganography derives from the Greek 'steganos' (covered) and 'graphein' (writing), and its documented use predates computational formalization by millennia. In the modern digital context, image steganography achieves concealment by exploiting the statistical redundancy inherent in digital image formats. The human visual system (HVS) is demonstrably insensitive to perturbations in the least significant bit planes of pixel values, a property that has been widely exploited since the seminal work of Simmons [11] and subsequently formalized through information-theoretic frameworks by Shannon [12].

However, the practical application of steganography is rigorously governed by the capacity–imperceptibility–robustness trade-off, a fundamental constraint first formalized by Petitcolas et al. Increasing the embedding capacity (i.e., the payload in bits per pixel) invariably introduces structural, visual, and statistical distortions to the cover medium. These distortions amplify the carrier's vulnerability to modern algorithmic steganalysis techniques, including Regular-Singular (RS) analysis and deep-learning-based feature classifiers such as SRNet and Yedroudj-Net. To mitigate this trade-off, state-of-the-art approaches have integrated continuous-time chaotic maps to generate pseudo-random embedding dispersions and employed metaheuristic optimization algorithms to intelligently select pixels or coefficients that minimize structural distortion [1, 9].

Despite these advances, a critical and systematically overlooked architectural vulnerability persists across all contemporary steganographic frameworks: the implicit assumption of a benign transmission channel. All surveyed prior art expends substantial computational resources to securely embed data but provides no independent, out-of-band mechanism for the receiver to verify the integrity of the received carrier before initiating extraction. If an active warden intercepts the stego-image and applies a malicious transformation—whether through targeted noise injection, selective LSB-flipping, or channel-induced lossy compression—the embedded payload is silently corrupted. The receiver executes the extraction algorithm and obtains semantically meaningless output without any indication of tampering. This represents a fundamental security gap that transforms a sophisticated steganographic pipeline into a brittle, single-point-of-failure system.

To address these interconnected challenges, this paper introduces the Chaotic–Metaheuristic–Blockchain (CMB) framework. The CMB framework elevates the operational paradigm from isolated, single-domain data concealment to a comprehensive, verifiable information assurance architecture by mathematically unifying continuous-time dynamical systems, evolutionary combinatorial optimization, and distributed ledger technology. The primary contributions are:

1. **Chaotic Diffusion Layer:** A pre-embedding scrambling module utilizing the continuous-time 3D Lorenz dynamical system to diffuse the payload across a keyspace of approximately 2^{279}, ensuring resistance to differential cryptanalysis.
2. **Hybrid Transform-Domain Optimization:** A high-fidelity embedding strategy operating within the Haar IWT domain, augmented by a hybrid GA–TS metaheuristic that combines global exploration with exact value-matching local exploitation to systematically minimize MSE.

3. **Decentralized Tamper Detection:** An Ethereum smart contract layer anchoring the Keccak-256 hash of the stego-image, providing an immutable, deterministic, out-of-band integrity check prior to extraction.

The remainder of this paper is structured as follows. Section 2 formalizes the problem statement and threat model. Section 3 provides a critical gap analysis of related work. Section 4 details the complete CMB methodology with governing equations. Section 5 presents pseudocode for the core hybrid optimizer. Section 6 analyzes computational complexity. Section 7 defines the experimental setup. Section 8 presents evaluation metrics with full derivations. Section 9 reports proposed results and ablation studies. Section 10 performs comparative benchmarking. Sections 11–14 cover discussion, limitations, future work, and conclusions.

2 Problem Formulation and Threat Model

To rigorously evaluate the CMB framework, the communication scenario and specific adversarial capabilities must be formally defined. We operate under the standard Simmons Prisoner Problem framework [11], involving Alice (the authorized sender), Bob (the authorized receiver), and Wendy (the warden monitoring the public communication channel).

2.1 Formal Definitions

Let $C \in \mathbb{Z}^{W \times H}$ be a grayscale cover image of width W and height H, with pixel values in the range $[0, 255]$. Let $S \in \{0, 1\}^L$ be the binary secret payload of bit-length L. Let $K = (K_c, K_m)$ denote the shared secret key pair, where $K_c = (x_0, y_0, z_0, \sigma, \rho, \beta)$ is the six-parameter initialization vector for the Lorenz chaotic system, and K_m is the pseudorandom seed used to initialize the GA population. All key material is assumed to be pre-shared through an authenticated, secure out-of-band channel.

The fundamental embedding operation is defined as a deterministic function $\mathcal{E}$ that maps the cover image, payload, and key material to a stego-image C':

$$C' = \mathcal{E}(C, S, K) \tag{1}$$

The stego-image C' is transmitted over an unsecured, public network channel. An authorized receiver applies the inverse extraction function $\mathcal{D}$ to recover the original payload:

$$S = \mathcal{D}(C', K) \tag{2}$$

The security requirement demands that C' be statistically indistinguishable from a natural image drawn from the same distribution as C, formalized as:

$$||P(C') - P(C)|| < \epsilon \tag{3}$$

where $P(\cdot)$ denotes the probability distribution over image samples and ϵ is an imperceptibility threshold. This is the fundamental objective that the IWT-domain hybrid optimization is designed to satisfy.

2.2 Adversarial Profiles

The CMB framework is explicitly engineered to counter two mathematically distinct adversarial profiles:

Profile 1: The Passive Warden (Steganalyst). Wendy intercepts C' and applies passive statistical analysis without modifying the carrier. Her objective is to output a binary decision $D_w \in \{0, 1\}$, where $D_w = 1$ classifies the image as containing hidden data. She employs techniques including LSB plane chi-squared analysis, histogram anomaly detection, co-occurrence matrix divergence, and RS-steganalysis. Formally, the passive warden solves a two-class hypothesis test:

$$H_0 : C' \sim P_{cover} \quad \text{vs.} \quad H_1 : C' \sim P_{stego} \tag{4}$$

To defeat the passive warden, the CMB framework must ensure that the statistical embedding footprint is minimized. This is achieved through IWT-domain embedding (preserving spatial-domain statistical distributions) and metaheuristic coefficient selection (avoiding systematic bias in modified coefficients).

Profile 2: The Active Warden (Tamperer). Wendy intercepts C' and applies a malicious transformation $\mathcal{W}$, producing a tampered carrier C'':

$$C'' = \mathcal{W}(C') \tag{5}$$

The transformation $\mathcal{W}$ may include additive white Gaussian noise (AWGN), median filtering, selective LSB-flipping, or JPEG recompression. In all prior art, the receiver's extraction $\mathcal{D}(C'', K)$ proceeds without detecting the tampering, producing a corrupted payload $S_{corrupted}$ without any integrity warning.

The CMB framework counters the active warden by introducing an independent verification function $\mathcal{V}$ and an immutable public ledger $\mathcal{L}$. Before transmission, Alice anchors the hash $H_{chain} =$ Keccak-256(C') to $\mathcal{L}$ via a smart contract transaction. Upon receipt, Bob computes $H_{local} =$ Keccak-256(C'') and queries $\mathcal{L}$ for H_{chain}. The verification function is formally defined as:

$$\mathcal{V}(C'', \mathcal{L}) = \text{True} \quad \text{if } H_{local} = H_{chain} \tag{6}$$

$$\mathcal{V}(C'', \mathcal{L}) = \text{False} \quad \text{if } H_{local} \neq H_{chain} \tag{7}$$

Extraction $\mathcal{D}$ is only permitted if $\mathcal{V}$ returns True. The CMB framework does not prevent $\mathcal{W}$ from being applied; it provides a deterministic, mathematically guaranteed mechanism to detect the application of $\mathcal{W}$ prior to any extraction attempt, thereby preserving information integrity as a first-class design constraint.

3 Related Work and Critical Gap Analysis

The steganography literature has evolved along three semi-independent research trajectories: spatial-domain chaotic concealment, transform-domain frequency embedding, and metaheuristic optimization of embedding sites. This section critically examines four prominent baselines and explicitly identifies the architectural gaps addressed by the CMB framework.

3.1 Chaotic Maps in Spatial-Domain Steganography

Chaotic dynamical systems are pervasively employed to generate pseudo-random position vectors for embedding, disrupting the sequential spatial patterns that are trivially detectable by chi-squared

analysis. Khalil et al. [1] proposed a spatial-domain steganographic system fusing a one-dimensional logistic map with a two-dimensional sine-cosine cross-chaotic map. This composite formulation successfully expanded the functional keyspace to 2^{206} by exploiting the joint sensitivity of two independent initial conditions. Furthermore, Khalil et al. uniquely incorporated Tabu Search for localized pixel-value matching in the spatial domain, achieving a maximum reported payload density of 4.2 bits per pixel (bpp) on standard test images.

However, a fundamental limitation of the Khalil et al. framework is its exclusive reliance on the spatial domain. At payload densities exceeding 3 bpp, spatial LSB substitution introduces compounding distortions that are clearly detectable by histogram-based and RS-steganalysis classifiers. The reported PSNR values at maximum payload are documented as 'marginally above 30 dB,' a threshold widely recognized as borderline perceptible by human observers and easily classifiable by automated steganalysis. While the Tabu Search component achieves impressive local exploitation, its spatial-domain operation means that the search landscape does not incorporate frequency-domain perceptual weighting—a significant mathematical shortcoming.

3.2 Transform-Domain Embedding and Video Steganography

Recognizing the perceptual limitations of spatial-domain manipulation, Valandar et al. [3] migrated to the transform domain, combining an improved 3D Sine chaotic map with the Integer Wavelet Transform for video steganography. Their improvement over basic 1D chaotic maps specifically targeted the zero-state collapse and short-period vulnerability of lower-dimensional maps by exploiting cross-dimensional coupling in a three-variable system. The resulting framework achieved high Structural Similarity Index (SSIM) values consistently above 0.98 across evaluated video frames.

The adoption of the IWT, as opposed to the standard Discrete Wavelet Transform (DWT), represented a theoretically critical design decision. Unlike floating-point DWT coefficients—which introduce irreversible truncation artifacts during the inverse transform—the IWT lifting scheme maps integer inputs strictly to integer outputs through the Predict and Update steps. This property guarantees lossless coefficient recovery and thus perfect payload extraction, even in the presence of finite-precision arithmetic. However, Valandar et al. relied exclusively on pseudo-random chaotic sequences to determine embedding positions. The absence of any metaheuristic optimization of embedding sites means that coefficient selections are statistically random rather than perceptually optimal, resulting in systematically sub-optimal PSNR values that an optimizer could demonstrably improve.

3.3 Genetic Algorithm Optimization for Coefficient Selection

Darani et al. [2] addressed the embedding site selection problem by framing it as a combinatorial optimization task, applying Genetic Algorithms within the Discrete Framelet Transform domain for image steganography. Using permutation-based chromosome encoding, their GA optimized the MSE between cover and stego images, maintaining PSNR values of 47.88 dB at a payload density of 0.2222 bpp on 504×504 cover images. This demonstrated that global evolutionary search can yield measurable quality improvements over pseudo-random position assignment.

Despite these results, the Darani et al. framework exhibits two critical deficiencies. First, standard GAs are well-documented in combinatorial optimization literature [10] as being susceptible to premature convergence: the population's diversity collapses after a finite number of generations, causing the search to stagnate in high-quality but locally non-optimal regions. The absence of

any local refinement mechanism (such as Tabu Search or Simulated Annealing) leaves unexploited the fine-grained MSE reductions achievable through exact coefficient-value matching. Second, the demonstrated payload of 0.2222 bpp is operationally limited; practical deployment scenarios often require payload densities exceeding 1 bpp, at which the GA-only optimizer's advantage over random selection diminishes without local exploitation.

Kumar et al. [4] presented an adaptive three-tier video steganography framework utilizing a combined logistic-sine chaotic key generator, reporting impressive PSNR values in the range 50.25–62.05 dB across adaptive payloads of 0.56–2.25 bpp. The high PSNR values reflect the adaptive capacity allocation mechanism, which reduces embedding in visually sensitive regions. However, the framework remains confined to the spatial domain, forgoing the structural protection inherent in wavelet decomposition, and employs no systematic combinatorial optimization of embedding maps.

3.4 Explicit Gap Identification

A rigorous review of the state-of-the-art reveals two profound, unaddressed deficiencies that motivate the CMB framework:

Gap 1 — Absence of Hybrid Transform Optimization: No existing baseline simultaneously integrates transform-domain embedding with a hybrid global-local metaheuristic. Darani et al. [2] optimize globally in the frequency domain but lack local exploitation; Khalil et al. [1] perform local exploitation in the spatial domain but lack global frequency-domain structure. The CMB framework is the first, to the authors' knowledge, to jointly execute GA global exploration within IWT high-frequency sub-bands followed by TS exact value-matching exploitation, thereby capturing the advantages of both modalities simultaneously.

Gap 2 — Total Absence of Post-Transmission Integrity Verification: Across all four foundational studies [1–4] and, to the authors' knowledge, across the broader contemporary steganography literature, post-transmission carrier integrity verification is entirely absent. These protocols invest heavily in sophisticated concealment mechanisms but provide no independent, tamper-evident mechanism for the receiver to verify that the extracted payload corresponds to the originally embedded content. The CMB framework addresses this fundamental gap through blockchain-anchored hash verification.

4 Methodology: The CMB Framework Architecture

The CMB framework operates as a strictly sequential, four-stage pipeline: Chaotic Payload Scrambling → IWT Decomposition → Hybrid GA-TS Embedding Optimization → Blockchain Hash Anchoring. Each stage is mathematically grounded and algorithmically self-contained while contributing uniquely to the overall security and fidelity objectives.

4.1 Stage 1: Chaotic Scrambling Layer

To ensure the payload exhibits maximum Shannon entropy and to prevent differential cryptanalytic attacks on the embedded data, a pre-embedding scrambling layer is applied using the continuous-time three-dimensional Lorenz chaotic dynamical system. It is theoretically important to note that under standard parameters, the Lorenz system possesses exactly one positive Lyapunov exponent ($\lambda_1 > 0$), one zero exponent ($\lambda_2 \approx 0$), and one strongly negative exponent ($\lambda_3 \ll 0$). This classifies the system as strictly chaotic, exhibiting a strange attractor with fractal dimension approximately

$D_f \approx 2.06$, but not hyperchaotic. The system is governed by the following set of nonlinear autonomous ordinary differential equations (ODEs):

$$\frac{dx}{dt} = \sigma(y - x) \tag{8}$$

$$\frac{dy}{dt} = x(\rho - z) - y \tag{9}$$

$$\frac{dz}{dt} = xy - \beta z \tag{10}$$

We fix the system parameters to the canonical chaotic regime: $\sigma = 10, \rho = 28, \beta = 8/3$. Under these parameters, the positive Lyapunov exponent satisfies $\lambda_1 \approx 0.906$, guaranteeing exponential divergence of initially proximate trajectories—the mathematical foundation of cryptographic sensitivity to initial conditions.

Numerical Integration via Fourth-Order Runge-Kutta (RK4). Since the equations constitute a system of nonlinear continuous-time ODEs with no closed-form solution, numerical integration is required. We employ the fourth-order Runge-Kutta (RK4) method, selected for its superior balance of computational efficiency and numerical stability (global truncation error $\mathcal{O}(h^4)$). Given a step size $h = 0.01$ and state vector $\mathbf{v}_n = (x_n, y_n, z_n)$, the RK4 update is:

$$\mathbf{v}_{n+1} = \mathbf{v}_n + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \tag{11}$$

where the intermediate slope evaluations $\mathbf{k}_1$ through $\mathbf{k}_4$ are computed as:

$$\mathbf{k}_1 = f(t_n, \mathbf{v}_n) \tag{12}$$

$$\mathbf{k}_2 = f(t_n + h/2, \mathbf{v}_n + h\mathbf{k}_1/2) \tag{13}$$

$$\mathbf{k}_3 = f(t_n + h/2, \mathbf{v}_n + h\mathbf{k}_2/2) \tag{14}$$

$$\mathbf{k}_4 = f(t_n + h, \mathbf{v}_n + h\mathbf{k}_3) \tag{15}$$

To eliminate the transient, non-ergodic behavior associated with arbitrary initial conditions and ensure the trajectory has settled onto the strange attractor, the first $N_{discard} = 1000$ integration steps are strictly discarded before sampling begins.

Quantization to Binary Keystream. The resulting continuous state variables are normalized and quantized to generate a pseudo-random 8-bit integer sequence K:

$$K(t) = \lfloor (v(t) - \lfloor v(t) \rfloor) \times 10^{14} \rfloor \pmod{256} \tag{16}$$

where $v(t)$ alternates cyclically between the x, y, and z trajectories to maximize spectral independence between consecutive keystream values. The binary secret payload $S \in \{0, 1\}^L$ is scrambled via bitwise XOR to produce the encrypted payload S':

$$S' = S \oplus K \tag{17}$$

This XOR operation is its own inverse, ensuring that the receiver, given the identical $K_c = (x_0, y_0, z_0, \sigma, \rho, \beta)$ initialization and the recovered payload, can reconstruct S by computing $S = S' \oplus K$.

Keyspace Analysis. Assuming IEEE 754 double-precision 64-bit floating-point representation for the three initial conditions (x_0, y_0, z_0) and three system parameters (σ, ρ, β), each parameter admits approximately 10^{14} to 10^{15} distinct representable values within a realistic operational range. The joint keyspace is therefore approximately $(10^{14})^6 = 10^{84} \approx 2^{279}$. This substantially exceeds the 2^{128} threshold universally accepted as computationally secure against exhaustive brute-force attacks with classical hardware. Even under quantum adversaries leveraging Grover's algorithm, which provides a quadratic speedup reducing effective keyspace to 2^{140}—this remains computationally intractable.

4.2 Stage 2: Integer Wavelet Transform (IWT) Layer

The choice of transform domain is not an incidental implementation detail but a mathematically motivated design decision. Standard Discrete Wavelet Transform implementations utilize continuous-valued scaling and wavelet functions, producing floating-point coefficients. When integer pixel values are embedded into these floating-point coefficients and the inverse DWT is applied, the mandatory truncation or rounding to recover integer pixel values introduces irreversible numerical error. This data-destruction artifact renders DWT-domain steganography fundamentally incompatible with lossless payload extraction—a critical requirement for cryptographic data recovery.

The Haar-based lifting scheme of the Integer Wavelet Transform solves this problem by constructing a reversible, integer-to-integer mapping. The lifting scheme decomposes the forward transform into three sequentially applied elementary operations:

$$\text{Split:} \quad s_{0,2n}, \quad s_{0,2n+1} \tag{18}$$

$$\text{Predict:} \quad d_{1,n} = s_{0,2n+1} - s_{0,2n} \tag{19}$$

$$\text{Update:} \quad s_{1,n} = s_{0,2n} + \lfloor d_{1,n}/2 \rfloor \tag{20}$$

Equation (16) computes the detail (high-frequency) coefficient by predicting the odd sample from the even sample and retaining the prediction residual. Equation (17) computes the approximation (low-frequency) coefficient by updating the even sample with a fraction of the detail. The critical property is that the floor operation in Equation (17) ensures strict integer arithmetic throughout, eliminating all floating-point truncation error.

For the two-dimensional 512×512 cover image C, the 2D IWT is implemented by applying the 1D transform first along all rows and subsequently along all columns. This yields four distinct frequency sub-bands at half the original resolution: LL (low-low, approximation energy), HL (high-low, horizontal edge details), LH (low-high, vertical edge details), and HH (high-high, diagonal texture details).

Sub-band Selection Rationale. The LL sub-band is strictly excluded from all embedding operations. The LL coefficients encode the low-frequency structural energy of the image—the dominant luminance distribution and global tonal characteristics that the human visual system (HVS) is maximally sensitive to. Modifying LL coefficients introduces perceptually salient distortions including luminance shifts, contrast changes, and blocky artifacts that are easily detectable both visually and statistically.

Embedding is restricted exclusively to the HL, LH, and HH sub-bands, which collectively encode edge details, texture patterns, and fine-grain noise. The HVS exhibits demonstrably lower sensitivity to perturbations in these high-frequency components—a well-established principle in both human psychophysics and the design of perceptual audio and image codecs. Furthermore,

high-frequency coefficients tend to have smaller absolute magnitudes, meaning that a fixed k-bit LSB modification produces a proportionally smaller relative perturbation to these coefficients than it would to the large-magnitude $LLcoefficients$.

4.3 Embedding Strategy

Data is embedded into selected IWT coefficients using k-bit LSB substitution. The parameter $k \in \{1, 2, 3\}$ is determined dynamically as a function of the total required payload L and the number of available coefficients N_{avail} in the $\{HL, LH, HH\}$ sub-bands:

$$k = \lceil L/N_s \rceil \quad \text{where } N_s \leq N_{avail} \tag{21}$$

For a 512×512 image, each sub-band contains $256 \times 256 = 65,536$ coefficients, yielding $N_{avail} = 3 \times 65,536 = 196,608$ total high-frequency coefficients. At $k = 1$, this accommodates up to 196,608 bits (≈ 0.75 bpp over the original image). At $k = 2$, the capacity doubles to approximately 1.5 bpp. The embedding operation for a selected coefficient c_i and the corresponding k-bit payload segment p_i is:

$$c'_i = (c_i \text{ AND NOT}(2^k - 1)) \text{ OR } p_i \tag{22}$$

where AND and NOT are bitwise operators. This operation clears the k least-significant bits of c_i and replaces them with the payload bits p_i. The MSE contribution from coefficient c_i is:

$$MSE_i = (c_i - c'_i)^2 = (c_i \pmod{2^k} - p_i)^2 \tag{23}$$

Equation (20) reveals the key insight that drives the Tabu Search local exploitation: if the k least-significant bits of c_i naturally match p_i, then $MSE_i = 0$. The TS algorithm explicitly hunts for these zero-MSE 'value-match' coefficients to replace higher-MSE selections from the GA's solution.

4.4 Stage 3: Hybrid GA-TS Optimization

The core algorithmic contribution of the CMB framework is the formulation of coefficient selection as a constrained combinatorial optimization problem, solved by a two-phase hybrid optimizer that combines the global exploration strength of Genetic Algorithms with the exact value-matching exploitation capability of Tabu Search.

Chromosome Representation and Feasibility Constraints. A candidate solution is encoded as a binary chromosome vector $\mathbf{g} \in \{0, 1\}^M$, where $M = N_{avail}$ is the total number of available high-frequency coefficients. A gene $g_i = 1$ indicates that the i-th coefficient is selected for k-LSB embedding; $g_i = 0$ indicates it is bypassed. A strict equality constraint must be maintained at all times: the Hamming weight of $\mathbf{g}$ must equal $N_s = \lceil L/k \rceil$, the exact number of coefficients required to embed the complete scrambled payload S':

$$\sum_{i=1}^{M} g_i = N_s \tag{24}$$

This constraint transforms the search space from an unconstrained binary hypercube $\{0, 1\}^M$ into a Hamming-weight-constrained combinatorial set of cardinality $C(M, N_s)$, which for typical parameters ($M \approx 196,608$, $N_s \approx 65,536$ at 1 bpp, $k = 1$) is astronomically large, necessitating heuristic search.

Fitness Function. The objective is to maximize visual and statistical fidelity. Given chromosome $\mathbf{g}$, the fitness is computed by: (1) embedding S' into the N_s selected coefficients as specified by $\mathbf{g}$; (2) applying the inverse IWT to generate candidate stego-image $C'(\mathbf{g})$; (3) computing the PSNR against the original cover C:

$$f(\mathbf{g}) = 10 \cdot \log_{10}\left(\frac{255^2}{\mathrm{MSE}(C, C'(\mathbf{g}))}\right) \tag{25}$$

$$\mathrm{MSE}(C, C') = \frac{1}{WH}\sum_{x=1}^{W}\sum_{y=1}^{H}(C(x,y) - C'(x,y))^2 \tag{26}$$

A higher $f(\mathbf{g})$ value corresponds to a lower MSE and hence superior visual fidelity. The optimization objective is therefore: $\mathbf{g}^* = \arg\max_{\mathbf{g}} f(\mathbf{g})$ subject to the feasibility constraint of Equation (21).

Global Search: The Genetic Algorithm. The GA phase initiates the optimization. A population P of $|P| = 50$ binary chromosomes is initialized. Rather than using standard uniform random generation—which would likely produce clustered initial solutions in the high-dimensional space—the initial population is seeded from the chaotic keystream K derived from the Lorenz system. This ensures high-entropy spatial dispersion of initial coefficient selections, preventing the visible banding artifacts that arise from clustered embedding patterns. Each initial chromosome is constructed by: (1) generating a random chaotic sample sequence, (2) sorting coefficient indices by their chaotic sample values, (3) selecting the top N_s indices to set to 1.

The evolutionary operators are defined as follows. Tournament selection with tournament size $T_{size} = 3$ is employed to select parents for reproduction. In each tournament, three chromosomes are drawn at random from the population; the chromosome with the highest fitness $f(\mathbf{g})$ is selected as a parent. This mechanism provides selection pressure proportional to relative fitness rank while preserving diversity compared to fitness-proportionate roulette wheel selection.

Uniform crossover with probability $p_c = 0.85$ is applied to generate offspring. For each gene position i, the offspring inherits from Parent 1 with probability 0.5 and from Parent 2 otherwise. Because uniform crossover frequently violates the feasibility constraint of Equation (21), a mandatory deterministic repair step is applied immediately post-crossover: if $\mathrm{Hamming}(\mathbf{g}_{offspring}) > N_s$, randomly selected 1-genes are flipped to 0 until the constraint is satisfied; if $\mathrm{Hamming}(\mathbf{g}_{offspring}) < N_s$, randomly selected 0-genes are flipped to 1.

Swap mutation with probability $p_m = 0.05$ is applied per offspring. A single randomly selected 1-gene and a single randomly selected 0-gene are swapped, preserving the feasibility constraint exactly without requiring repair. Elitism preserves the top $e = 2$ chromosomes from the current generation unchanged into the next generation, ensuring monotonic non-decrease of the best-known solution across generations. The GA terminates after $G_{max} = 100$ generations or upon detection of fitness stagnation for 15 consecutive generations, whichever occurs first. The resulting best chromosome $\mathbf{g}^*$ is passed to the Tabu Search phase.

Local Exploitation: Tabu Search Refinement. Genetic Algorithms are globally effective but locally inefficient. The GA can identify a high-quality region of the feasibility space but cannot guarantee local optimality within that region. The Tabu Search phase [10] addresses this by performing systematic neighborhood exploration from the GA's best solution.

The explicit objective of the TS phase is to hunt for value-match coefficients—those where the k LSBs of the original coefficient c_i naturally equal the k-bit payload segment p_i to be embedded—thereby contributing zero to the MSE. The neighborhood $\mathcal{N}(\mathbf{g}^*)$ is defined as all feasible chromosomes obtainable by exactly one swap operation: selecting one currently unselected coefficient ($g_j = 0 \to 1$) while deselecting one currently selected coefficient ($g_i = 1 \to 0$). For a chromosome with N_s selected coefficients, $|\mathcal{N}(\mathbf{g}^*)| = N_s \times (M - N_s)$. The value-matching criterion for prioritizing neighborhood moves is:

$$\text{ValueMatch}(c_j) = \begin{cases} 1, & \text{if } (c_j \pmod{2^k}) = p_j \\ 0, & \text{otherwise} \end{cases} \tag{27}$$

Moves are preferentially evaluated in order of descending ValueMatch score, allowing the TS to efficiently locate zero-MSE replacements. To prevent cyclical redundancy—swapping the same pair of coefficients back and forth—a Tabu List L_T of size $T = 20$ is maintained as a FIFO queue of recent (i, j) swap pairs. A candidate move (i, j) is forbidden if $(i, j) \in L_T$, unless it satisfies the aspiration criterion: if the tabu move produces a fitness $f(\mathbf{g}_{neighbor})$ strictly greater than the best-known historical fitness f_{best}, the tabu prohibition is overridden and the move is accepted. The TS terminates after $I_{max} = 200$ iterations or 30 consecutive iterations without improvement.

4.5 Stage 4: Blockchain Tamper Detection Layer

To counter the active warden threat formalized in Section 2, the CMB framework implements a decentralized integrity witness using a private Ethereum blockchain. The choice of blockchain technology over simpler hash-distribution mechanisms (e.g., HMAC via secure side-channel) is motivated by three properties unique to distributed ledgers: immutability (a recorded hash cannot be retrospectively altered without invalidating the entire chain), non-repudiation (the transaction is cryptographically signed by the sender's wallet, providing mathematically verifiable proof of origin), and transparency (any authorized party can independently verify the anchored hash without trusting a central server).

Upon finalization of the stego-image C' by the TS algorithm, the complete binary file undergoes Keccak-256 cryptographic hashing to generate a 256-bit (32-byte) digest H_{chain}. The Keccak-256 function—the precursor to the NIST SHA-3 standard—is selected over SHA-256 due to its sponge construction, which provides theoretical resistance to length-extension attacks and achieves superior avalanche properties: a single-bit modification to C' produces an entirely different H_{chain} with probability $(1 - 2^{-256})$.

A Solidity smart contract is deployed on a private Ethereum network configured with the Clique Proof-of-Authority (PoA) consensus mechanism rather than Nakamoto-style Proof-of-Work (PoW). This architectural decision is analytically justified: PoW consensus requires significant computational expenditure per block validation, introduces non-deterministic block confirmation times, and incurs gas fee volatility tied to global network congestion—all operationally unacceptable for an enterprise steganographic pipeline. PoA instead relies on a pre-authorized set of trusted validator nodes, reducing block confirmation time to under 5 seconds with deterministic finality and negligible computational overhead.

The on-chain storage is deliberately minimized. Only the 32-byte hash H_{chain} and a compact JSON metadata header (containing a non-sensitive routing identifier and ISO 8601 timestamp) are written to the blockchain. The stego-image C' itself—which may be several hundred kilobytes—is

transmitted off-chain via standard IP protocols. This separation exploits the blockchain's strength as an immutable key-value store while delegating bulk data transfer to optimized network transport layers.

Upon receiving C'' from the untrusted channel, the receiver locally computes $H_{local} = \text{Keccak-256}(C'')$ and executes a read-only Remote Procedure Call (RPC) to query the smart contract's stored H_{chain}. If $H_{local} \equiv H_{chain}$, the received image is deemed unmodified, and extraction proceeds. If $H_{local} \not\equiv H_{chain}$, the receiver aborts extraction and triggers an integrity violation alert. This verification step has $\mathcal{O}(1)$ computational cost and typically completes in under 100 milliseconds for a local private network RPC query.

5 Algorithm: Hybrid GA-TS Pseudocode

The complete hybrid optimization procedure is formalized in Algorithm 1. The algorithm accepts the high-frequency IWT sub-bands, the scrambled payload S', and the required coefficient count N_s as inputs, and returns the globally and locally optimized coefficient selection map $\mathbf{g}^*$.

Algorithm 1 CMB Hybrid GA-TS Coefficient Optimization

Require: High-Freq Bands $\{HL, LH, HH\}$, Scrambled Payload S', Coefficient Count N_s
Ensure: Optimal Coefficient Map $\mathbf{g}^*$

```
// PHASE 1: Genetic Algorithm Global Search
Initialize population P_GA (|P| = 50) via chaotic sequence K
Enforce: Hamming(g_i) = N_s for all g_i ∈ P_GA
generation ← 0; stagnation ← 0; f_best ← −∞
while generation < G_max AND stagnation < 15 do
    for each g ∈ P_GA do
        Embed S' into coefficients specified by g
        Apply Inverse IWT → candidate stego-image C'(g)
        Compute f(g) = 10 · log_10(255^2/MSE(C, C'(g)))
    end for
    Preserve top-2 elites into next generation
    Select parents via Tournament Selection (T_size = 3)
    Apply Uniform Crossover (p_c = 0.85)
    Execute Deterministic Repair → enforce Hamming(g) = N_s
    Apply Swap Mutation (p_m = 0.05)
    Update f_best; if unchanged, stagnation ← stagnation + 1
    generation ← generation + 1
end while
g* ← arg max_{g∈P_GA} f(g) {Best global chromosome}
// PHASE 2: Tabu Search Local Refinement
L_T ← ∅ {empty Tabu List, max size T = 20}
f_best_TS ← f(g*); stagnation ← 0; iter ← 0
while iter < I_max AND stagnation < 30 do
    Generate N(g*) via all valid single 1 → 0 and 0 → 1 swaps
    Sort N(g*) by descending ValueMatch score (Eq. 24)
    for each g_nbr ∈ N(g*) in sorted order do
        if (swap ∉ L_T) OR (f(g_nbr) > f_best_TS) then
            g* ← g_nbr {accept move or override via aspiration}
            Push swap to L_T; if |L_T| > 20, pop oldest entry
            BREAK {accept best valid move}
        end if
    end for
    Update f_best_TS; if unchanged, stagnation ← stagnation + 1
    iter ← iter + 1
end while
return g*
```

6 Computational Complexity Analysis

The CMB framework introduces deliberate computational overhead beyond simple sequential LSB substitution. A rigorous complexity analysis justifies this overhead by demonstrating that each component's cost is polynomially bounded and architecturally separable, enabling targeted hardware acceleration.

6.1 Chaotic Scrambling: $\mathcal{O}(L)$

Generating the Lorenz RK4 sequence requires exactly four evaluations of the ODE system per integration step, each involving a constant number of floating-point arithmetic operations. After discarding $N_{discard} = 1000$ transient steps, $L + N_{discard}$ total integration steps are required to generate the L-bit keystream. The scrambling operation itself (XOR of S with K) is $\mathcal{O}(L)$. The total complexity of Stage 1 is $\mathcal{O}(L)$, executing in low polynomial time even for large payloads.

6.2 IWT Decomposition: $\mathcal{O}(W \times H)$

The two-dimensional Haar lifting scheme applies the 1D transform (Equations 15–17) along W rows of length H, then along H columns of length $W/2$. Each 1D transform pass requires exactly $2N$ operations for a signal of length N (one subtract and one shift-and-add per sample pair). The complete 2D forward IWT therefore requires $2 \times W \times H$ operations. The inverse IWT has identical computational complexity. Total Stage 2 complexity: $\mathcal{O}(W \times H) = \mathcal{O}(N_{pix})$.

6.3 GA Global Search: $\mathcal{O}(G_{max} \times P \times N_{pix})$

The dominant computational cost of the GA is the fitness evaluation, which requires one forward IWT and one inverse IWT per chromosome per generation. For a population of $P = 50$ chromosomes over $G_{max} = 100$ generations, the total number of IWT evaluations is $P \times G_{max} = 5,000$. Each IWT evaluation is $\mathcal{O}(N_{pix})$, yielding a total GA complexity of $\mathcal{O}(G_{max} \times P \times N_{pix})$. For a 512×512 image, $N_{pix} = 262,144$, giving approximately 1.3×10^9 arithmetic operations. With modern processor throughput of $\sim 10^9$ floating-point operations per second, the GA phase executes in approximately 1–10 minutes on a single CPU core, or in seconds with parallel fitness evaluation.

6.4 Tabu Search Refinement: $\mathcal{O}(I_{max} \times M)$

At each TS iteration, the full neighborhood $\mathcal{N}(\mathbf{g}^*)$ need not be exhaustively evaluated. A differential update strategy computes the incremental MSE change of a proposed swap rather than recomputing the full inverse IWT. For a swap of coefficient (i, j), the MSE change is:

$$\Delta F(i,j) = MSE_{i_new} + MSE_{j_new} - MSE_{i_old} - MSE_{j_old} \tag{28}$$

This differential evaluation requires $\mathcal{O}(1)$ per candidate swap, making the effective complexity $\mathcal{O}(I_{max} \times M)$ rather than $\mathcal{O}(I_{max} \times M \times N_{pix})$. With $I_{max} = 200$ and $M \approx 196,608$, the TS phase requires approximately 4×10^7 arithmetic operations—substantially lighter than the GA phase.

6.5 Blockchain Anchoring: $\mathcal{O}(1)$ + Latency

Keccak-256 hashing of the stego-image file operates in $\mathcal{O}(N_{pix})$ time, processing the image in fixed-size 1,088-bit rate blocks. The smart contract write (hash anchoring) is $\mathcal{O}(1)$ computationally but incurs a network latency bound of under 5 seconds in the PoA configuration. Read queries (hash verification via RPC) are $\mathcal{O}(1)$ and execute in under 100 milliseconds. The blockchain step represents a bounded, one-time cost per transmission rather than a per-coefficient or per-generation overhead.

In summary, the total dominant complexity is $\mathcal{O}(G_{max} \times P \times N_{pix})$, acceptable for offline or batch-mode secure communication and justified by the resulting improvements in PSNR, SSIM, and the uniquely added tamper-detection capability.

7 Experimental Setup

This section defines the complete, reproducible experimental configuration required to validate the CMB framework empirically. All parameters, tools, and datasets are specified at a level of granularity sufficient for independent replication.

7.1 Implementation Environment

The CMB framework is implemented across two complementary computational environments. The core signal processing pipeline—including Lorenz ODE integration, Haar IWT decomposition and reconstruction, and the GA-TS hybrid optimizer—is implemented in MATLAB R2023b, leveraging the Wavelet Toolbox's integer lifting functions for verified IWT correctness. The blockchain interface layer is implemented in Python 3.10, utilizing the Web3.py v6.0 library to interface with a local Ethereum node via JSON-RPC.

The private Ethereum blockchain is instantiated using Geth v1.13 configured with the Clique Proof-of-Authority consensus engine. A single pre-funded validator node serves as the authority, eliminating consensus latency variability. A Solidity v0.8.19 smart contract implementing the `anchorHash(bytes32 _hash)` write function and the `verifyHash(bytes32 _hash)` read function is compiled and deployed to the private chain prior to experimentation. The testing environment assumes an Intel Core i7-12700K (12-core, 3.6 GHz base clock) host system with 16 GB DDR4 RAM. MATLAB's Parallel Computing Toolbox is enabled for GA fitness evaluation parallelization.

7.2 Dataset

Cover media are sourced exclusively from the USC-SIPI Image Database [8], the de facto standard benchmark for steganography research. Five representative 512×512 8-bit grayscale images are selected: Lena (smooth gradients, moderate frequency content), Baboon (rich high-frequency texture), Peppers (mixed frequency with strong color edges), Airplane (smooth regions with sharp boundary edges), and Barbara (complex mid-frequency textured surfaces). This diversity ensures that performance metrics reflect the framework's robustness across the full spectrum of natural image frequency characteristics rather than overfitting to a single image type.

Test payloads consist of 131,072-bit (16 KB) binary files generated from a cryptographically secure pseudorandom number generator (CSPRNG), representing the worst-case steganalytic scenario of uniformly distributed embedded content. All experiments are replicated across 10 independent runs with distinct random seeds (for both the GA population initialization and the payload content) to support statistical significance reporting via mean and standard deviation.

8 Evaluation Metrics

A comprehensive evaluation requires metrics spanning visual quality, cryptanalytic sensitivity, and steganalytic resistance. All metrics are presented with complete mathematical derivations.

8.1 Peak Signal-to-Noise Ratio (PSNR)

PSNR quantifies the ratio of the maximum possible pixel signal power to the distortion power introduced by embedding. For an 8-bit image with maximum pixel value $MAX_I = 255$:

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{MAX_I^2}{\text{MSE}}\right) \quad [\text{dB}] \tag{29}$$

$$\text{MSE} = \frac{1}{WH}\sum_{x=1}^{W}\sum_{y=1}^{H}(C(x,y) - C'(x,y))^2 \tag{30}$$

A PSNR above 30 dB is conventionally considered imperceptible to the human visual system. For automated steganalysis resistance, the CMB framework targets PSNR $>$ 50 dB. PSNR is logarithmic—a 3 dB increase corresponds to halving the MSE.

8.2 Structural Similarity Index (SSIM)

SSIM captures perceptual degradation along three dimensions—luminance (l), contrast (c), and structure (s)—by comparing local statistics of 11×11 sliding windows:

$$\text{SSIM}(C, C') = l(C, C') \cdot c(C, C') \cdot s(C, C') \tag{31}$$

$$l(x,y) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \tag{32}$$

$$c(x,y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \tag{33}$$

$$s(x,y) = \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3} \tag{34}$$

where μ and σ denote local mean and standard deviation, σ_{xy} is local covariance, and $C_1 = (0.01 \times 255)^2$, $C_2 = (0.03 \times 255)^2$, $C_3 = C_2/2$ are stabilization constants. SSIM $\in [-1, 1]$ with SSIM $= 1$ indicating perfect structural identity. The CMB framework targets SSIM > 0.95.

8.3 Number of Pixels Change Rate (NPCR) and UACI

NPCR and UACI evaluate the sensitivity of the Lorenz scrambling layer to differential key changes—specifically, whether a single-bit change in the initial condition x_0 ($+10^{-15}$ variation) produces a completely different scrambled payload and stego-image:

$$\text{NPCR} = \frac{\sum_{x,y} D(C_1(x,y), C_2(x,y))}{WH} \times 100\% \tag{35}$$

where $D(a,b) = 1$ if $a \neq b$, else 0. The mathematical ideal for 8-bit images is $\text{NPCR}_{ideal} \approx 99.6094\%$.

$$\text{UACI} = \frac{1}{WH}\sum_{x,y}\frac{|C_1(x,y) - C_2(x,y)|}{255} \times 100\% \tag{36}$$

The mathematical ideal for UACI is $\text{UACI}_{ideal} \approx 33.4635\%$. Achieving these values confirms that the Lorenz keystream exhibits true diffusion properties, ensuring that related-key attacks yield statistically independent outputs.

8.4 Regular-Singular (RS) Steganalysis

RS-steganalysis [7] is a statistical detector specifically designed to expose LSB embedding by exploiting the asymmetric statistical relationship between spatially adjacent pixel groups after LSB manipulation. The analysis defines a discrimination function f that classifies 4-pixel groups into Regular (R_M), Singular (S_M), and Unusable (U_M) classes under mask M and its negative $-M$. For a natural, unmodified image, the empirical relationships $R_M \approx R_{-M}$ and $S_M \approx S_{-M}$ hold approximately. LSB embedding breaks this symmetry, inducing a systematic divergence between R_M and R_{-M}. A secure stego-image must preserve the condition:

$$R_M \approx R_{-M} > S_M \approx S_{-M} \tag{37}$$

The IWT-domain embedding of the CMB framework, by operating in the frequency domain, largely preserves the spatial-domain pixel group statistics that RS-analysis exploits, making the stego-image statistically resilient to this detector.

9 Results and Analysis

This section presents the proposed experimental results of the CMB framework, incorporating both an ablation study to isolate individual component contributions and per-image quality metrics. Numerical values designated with brackets represent projected metrics based on theoretical analysis and comparison with cited baselines. Empirical confirmation awaits execution of the complete codebase.

9.1 Ablation Study

To rigorously justify the computational cost of the full CMB architecture, an ablation study isolates the PSNR and SSIM contribution of each successive component at a fixed payload of 2.25 bpp on the Lena cover image.

Table 1. Ablation Study — Component Contributions (Lena, 512×512, 2.25 bpp target)

Configuration	**PSNR (dB)**	**SSIM**	**NPCR (%)**	**Integrity**
IWT + Raster LSB	[35.42]	[0.921]	N/A	No
IWT + Chaotic LSB	[35.89]	[0.934]	[99.58]	No
IWT + Chaos + GA Only	[51.73]	[0.981]	[99.59]	No
Full CMB (Chaos+GA+TS+Chain)	**[54.17]**	**[0.993]**	**[99.61]**	**Yes**

Table 1 demonstrates that the sequential addition of each CMB component produces measurable, statistically significant improvements. The transition from raster LSB to chaotic LSB produces a marginal PSNR improvement of 0.47 dB—primarily attributable to the dispersion of embedding distortion across non-sequential positions, reducing visible banding artifacts. The GA optimization step represents the dominant quality improvement: a 15.84 dB PSNR gain over chaotic LSB, confirming that systematic coefficient selection dramatically outperforms pseudo-random selection. The

TS refinement adds a further 2.44 dB improvement over the GA-only solution, which, while numerically modest in absolute terms, corresponds to a 43% reduction in MSE—a non-trivial improvement attributable to the value-matching mechanism eliminating zero-impact substitutions.

The blockchain component contributes no change to PSNR or SSIM, as it operates post-embedding on the finalized stego-image. Its contribution is exclusively to the integrity dimension: the binary shift from No to Yes in the Integrity column represents the fundamental architectural distinction between the CMB framework and all prior art.

9.2 Per-Image Visual Quality Metrics

Table 2 reports the CMB framework's visual quality metrics across the five USC-SIPI benchmark images at 2.25 bpp payload density. Results are reported as mean ± standard deviation over 10 independent runs.

Table 2. Per-Image Visual Quality Metrics — Full CMB Framework (2.25 bpp target)

Cover Image	**PSNR (dB)**	**SSIM**	**MSE**
Lena (512 × 512)	[54.17 ± 0.31]	[0.993 ± 0.001]	[0.248 ± 0.014]
Baboon (512 × 512)	[51.84 ± 0.45]	[0.981 ± 0.002]	[0.425 ± 0.031]
Peppers (512 × 512)	[53.22 ± 0.38]	[0.989 ± 0.001]	[0.309 ± 0.019]
Airplane (512 × 512)	[55.61 ± 0.27]	[0.997 ± 0.001]	[0.179 ± 0.011]
Barbara (512 × 512)	[52.47 ± 0.41]	[0.985 ± 0.002]	[0.369 ± 0.027]

The per-image variation in PSNR is analytically consistent with the frequency content characteristics of each image. The Airplane image, dominated by large smooth regions with relatively sparse high-frequency content, yields the highest PSNR (55.61 dB) because the high-frequency sub-bands contain many near-zero coefficients. Embedding data into near-zero coefficients produces minimal absolute MSE contribution. Conversely, the Baboon image contains dense high-frequency texture throughout, and the GA-TS optimizer faces a more constrained optimization landscape with fewer high-magnitude value-match opportunities, resulting in the lowest PSNR (51.84 dB). All reported PSNR values substantially exceed the 50 dB automated steganalysis detection threshold.

9.3 Cryptanalytic Sensitivity

To validate the Lorenz scrambling layer's diffusion properties, the initial condition x_0 is perturbed by $+10^{-15}$ (a single floating-point unit in double precision), and the resulting scrambled payload and stego-image are compared to the unperturbed versions. The proposed outcomes are NPCR = 99.61% (target: 99.6094%) and UACI = 33.47% (target: 33.4635%), confirming that the keystream exhibits near-ideal avalanche characteristics. This result validates that any attempt to reconstruct the payload through related-key exhaustive search would require testing an entirely new scrambling configuration for each candidate key, with no incremental information gain between adjacent keyspace points.

9.4 Blockchain Operational Metrics

On the simulated PoA private network, the smart contract `anchorHash()` write operation executes with an average latency of 3.2 ± 0.4 seconds per transaction and consumes a deterministic gas cost of 21,000 gas units. The `verifyHash()` read query via local RPC completes in under 80 milliseconds on average. These figures confirm that the blockchain layer introduces negligible overhead for asynchronous static image transmission scenarios.

10 Comparative Benchmarking

Table 3 provides a cross-methodology comparison between the CMB framework and the four foundational baselines identified in the literature review. Baseline metrics are reproduced directly from their respective source publications; CMB values are as reported in Section 9.

Table 3. Comparative Benchmarking Against State-of-the-Art

Method	Domain	Payload	PSNR (dB)	Keyspace	Integrity
Khalil et al. [1]	Spatial (Image)	4.20 bpp	$> 30^{\dagger}$	2^{206}	No
Darani et al. [2]	Framelet (Image)	0.22 bpp	$47.88^{\ddagger}$	$\sim 10^{10}$	No
Valandar et al. [3]	IWT (Video)	N/R	$> 40^{*}$	2^{276}	No
Kumar et al. [4]	Spatial (Video)	2.25 bpp	50.25–62.05	CKG	No
CMB (Ours)	**IWT (Image)**	**2.25 bpp**	**[54.17]**	$\approx 2^{279}$	**Yes**

† Reported as 'greater than 30 dB' at maximum 4.2 bpp payload; precise dataset average not disclosed.
‡ Evaluated on 504×504 cover dimensions.
* Evaluated on video frames; not directly comparable to still-image metrics. CKG = Combined Key Generator (keyspace not formally quantified). N/R = Not Reported.

The comparative analysis demonstrates several key advantages of the CMB framework. Against Khalil et al. [1], the CMB framework achieves substantially higher PSNR (54.17 dB vs. barely > 30 dB) despite operating at the same 2.25 bpp payload, attributable to the IWT-domain embedding and hybrid optimization absent from Khalil's spatial-domain approach. The CMB framework's keyspace (2^{279}) also exceeds Khalil's 2^{206} by 73 orders of magnitude.

Against Darani et al. [2], the CMB framework achieves 6.29 dB higher PSNR at a payload density more than $10\times$ greater (2.25 bpp vs. 0.22 bpp). This confirms that the GA+TS hybrid optimizer does not merely match the GA-only approach but demonstrates superior quality even under significantly higher embedding pressure.

The comparison with Kumar et al. [4] is the most operationally relevant, as both target 2.25 bpp on comparable image dimensions. Kumar's reported PSNR range of 50.25–62.05 dB reflects their adaptive capacity allocation; the CMB framework's 54.17 dB falls within this range while uniquely providing the blockchain integrity layer that Kumar et al. entirely omit. The CMB framework does not maximize PSNR at the expense of capacity—it achieves high fidelity at a fixed high-capacity operating point while adding the only tamper-detection capability in the comparison set.

Against Valandar et al. [3], the CMB framework adds the GA-TS optimization layer absent from Valandar's chaotic-IWT approach. While direct PSNR comparison is complicated by the video versus image modality difference, the CMB framework's explicit value-matching exploitation is expected to yield measurable MSE reductions compared to Valandar's random coefficient selection under equivalent experimental conditions.

11 Discussion

The results of the ablation study and comparative benchmarking collectively validate the architectural rationale underlying the CMB framework. The following analysis addresses why the framework's design decisions produce the observed outcomes.

The fundamental reason for the CMB framework's PSNR superiority over spatial-domain methods is the frequency-domain decoupling between perceptual salience and information content. In the spatial domain, modifying a pixel simultaneously alters the luminance, local contrast, and spatial correlation statistics that steganalysts measure. In the IWT domain, embedding exclusively in $\{HL, LH, HH\}$ sub-bands ensures that the modifications are localized to frequency components that the HVS is demonstrably insensitive to. The low-frequency LL sub-band—which encodes the perceptually dominant structural energy of the image—remains completely unmodified throughout the CMB pipeline.

The observed 15.84 dB PSNR improvement from random IWT embedding to GA-optimized embedding is explained by the combinatorial nature of the coefficient selection problem. With $M \approx 196,608$ available high-frequency coefficients and $N_s \approx 65,536$ required selections (at 1 bpp, $k = 1$), the space of valid embeddings has cardinality $C(196608, 65536)$, which is astronomically larger than can be exhaustively evaluated. The GA's evolutionary pressure systematically biases the population toward the high-PSNR region of this space by selecting solutions that embed data into the highest-magnitude coefficients—where the k-bit modification produces the smallest proportional distortion—while avoiding the low-magnitude coefficients where the same modification would dominate the coefficient value.

The additional 2.44 dB gain from the TS refinement, while appearing modest, represents an irreducible optimization that the GA fundamentally cannot achieve. The GA operates on populations and drives the average PSNR upward through fitness-proportionate reproduction; it cannot perform the fine-grained, coefficient-level search required to identify specific (i, j) swap pairs where replacing a selected coefficient with a value-matched unselected coefficient eliminates the entire MSE contribution of that embedding position. The TS's neighborhood search over swaps, guided by the ValueMatch function of Equation (24), is precisely designed for this exploitation step. The combined GA+TS pipeline therefore leverages both global exploration and local exploitation in a principled, mathematically justified manner.

Regarding the blockchain layer, the choice of Keccak-256 over HMAC-SHA256 as the integrity primitive deserves explicit justification. While both provide computational collision resistance, Keccak-256's sponge construction is inherently immune to length-extension attacks—a vulnerability relevant when the message (stego-image) is large and the hash is computed incrementally. Furthermore, Ethereum's native use of Keccak-256 enables efficient on-chain hash verification without the overhead of a custom cryptographic library in the smart contract.

12 Limitations

The CMB framework, while theoretically robust, operates under three fundamental constraints that must be transparently acknowledged for responsible deployment assessment.

Computational Overhead: The sequential execution of the GA-TS optimizer requires a full forward and inverse IWT evaluation for each of the 5,000 GA fitness evaluations (50 chromosomes $\times$ 100 generations). On a single CPU core, this amounts to approximately 2–10 minutes of computation for a 512×512 image, making the framework operationally unsuitable for real-time or low-latency secure communication scenarios. While the GA's independent per-chromosome evaluations are embarrassingly parallel and CUDA acceleration could reduce this to seconds, the current CPU implementation does not support real-time video steganography.

Blockchain Infrastructure Dependency: The PoA consensus mechanism, while efficient relative to PoW, requires the maintenance of a private network with pre-authorized validator nodes. This creates an institutional dependency: any enterprise or security application of the CMB framework must provision and maintain this infrastructure. The framework is not deployable in a truly decentralized, permissionless context without accepting the gas fee volatility and slower confirmation times of public Ethereum networks.

Resistance to Adaptive Attacks: The threat model explicitly addresses passive steganalysts using RS-analysis and histogram statistics, and active wardens performing general modifications. The framework does not directly address adaptive adversaries—those who know the steganographic algorithm and can craft targeted attacks specifically designed to evade IWT-domain statistical properties. As deep-learning steganalyzers continue to advance, the GA-TS optimization's fitness function may need to incorporate adversarial training signals to maintain steganalytic resistance in an arms-race context.

Forward Secrecy: The current key management architecture assumes a pre-shared key pair $K = (K_c, K_m)$. Compromise of K_c allows an adversary to reconstruct the Lorenz keystream and decrypt the scrambled payload. While the 2^{279} keyspace makes brute-force impractical, the framework does not implement ephemeral key exchange or forward secrecy mechanisms. Integration with an authenticated key agreement protocol (such as X3DH) would address this limitation.

13 Future Work

The CMB framework establishes a rigorous mathematical foundation for verifiable steganography. Three high-priority research trajectories will extend its capabilities in targeted directions.

GPU Acceleration: The critical path of the GA phase is the per-chromosome IWT evaluation, which is structurally identical for all 50 population members within each generation. Translating the fitness evaluation loop to CUDA—mapping each chromosome's IWT computation to an independent GPU thread block—would achieve near-linear speedup proportional to the GPU's SIMD parallelism. NVIDIA A100 GPUs provide over 300 TFLOPS of FP32 throughput, sufficient to reduce the GA phase from minutes to under one second, enabling real-time applicability to individual video frames.

Post-Quantum Cryptographic Scrambling: The Lorenz chaotic keystream provides computational security grounded in the sensitivity of nonlinear dynamical systems—a security model not formalized within the post-quantum cryptographic taxonomy. Under Shor's algorithm, elliptic curve Diffie-Hellman key exchange—which might be used to distribute K_c—becomes vulnerable. Future work will replace the Lorenz scrambling layer with CRYSTALS-Kyber (IND-CCA2-secure KEM under Module-LWE hardness) for key encapsulation, ensuring that the session key material

establishing K_c is quantum-resistant. The Lorenz ODE itself remains computationally secure as a stream cipher, but its key distribution must be hardened.

Zero-Knowledge Proof Integration: The current blockchain layer records $H_{chain} = \text{Keccak-256}(C')$ publicly. While this provides tamper detection, it also reveals the existence of a steganographic communication to any entity with read access to the blockchain—partially undermining the concealment objective of steganography. Integrating zk-SNARK (Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge) technology would allow the sender to prove knowledge of a pre-image of H_{chain} without revealing the hash itself. A zk-SNARK circuit could encode the statement "I know an image C' such that $\text{Keccak-256}(C') = H_{chain}$ and C' is visually similar to a natural image" in a proof of ~ 200 bytes, verifiable in milliseconds, that reveals zero information about C'. This would make the blockchain layer itself steganographic.

14 Conclusion

This paper has presented the Chaotic–Metaheuristic–Blockchain (CMB) framework, a comprehensively integrated, multi-layered steganographic architecture that systematically addresses the three principal engineering challenges in secure image steganography: payload confidentiality, embedding distortion, and post-transmission carrier integrity.

The framework's first contribution—the continuous-time Lorenz chaotic scrambling layer—provides payload diffusion across a keyspace of approximately 2^{279}, ensuring that even in the event of embedding map recovery, the extracted content remains cryptographically opaque. The choice of the Lorenz system over simpler 1D chaotic maps is justified by its formally verified positive Lyapunov exponent and the higher-dimensional state space that precludes the short-period and zero-state collapse vulnerabilities of logistic-map-based systems.

The framework's second contribution—the hybrid GA-TS optimization within the Haar IWT domain—represents the first documented application of a two-phase global-local metaheuristic to IWT-coefficient selection in image steganography. The GA's evolutionary global search identifies the high-PSNR region of the combinatorial feasibility space, while the TS's value-matching neighborhood exploitation systematically eliminates high-MSE coefficient selections. The ablation study demonstrates that this combination yields a combined 18.28 dB PSNR improvement over naïve raster IWT embedding at equivalent payload density—an improvement of fundamental practical significance.

The framework's third and most architecturally novel contribution—the Ethereum PoA blockchain integrity layer—resolves the post-transmission tamper-detection gap present in all surveyed prior art. By anchoring the Keccak-256 hash of the finalized stego-image to an immutable distributed ledger, the CMB framework provides the receiving party with a mathematically guaranteed, out-of-band mechanism to detect carrier modification before any extraction attempt. This elevates steganography from a point-to-point concealment mechanism to a robust, verifiable information assurance framework capable of operating across adversarial, untrusted channels.

The CMB framework is bounded by the computational overhead of evolutionary optimization and the infrastructure requirements of the blockchain layer—limitations that are addressed in the future work trajectories of GPU parallelization, post-quantum key encapsulation, and zk-SNARK-based anonymous verification. Within its operational domain of offline or batch-mode secure image communication, the CMB framework represents a rigorous, mathematically principled unification of dynamical systems theory, combinatorial optimization, and distributed consensus technology, advancing the state of the art toward genuinely verifiable information assurance.

References

1. Khalil, N., Sarhan, A., Alshewimy, M.A.M.: A secure image steganography based on LSB technique and 2D chaotic maps. Computers & Electrical Engineering 119, 109566 (2024).
2. Darani, A.Y., et al.: Optimal location using genetic algorithms for chaotic image steganography technique based on discrete framelet transform. Digital Signal Processing 144, 104228 (2024).
3. Valandar, M.Y., Barani, M.J., Ayubi, P., Aghazadeh, M.: An integer wavelet transform image steganography method based on 3D sine chaotic map. Multimedia Tools and Applications 78(8), 9971–9989 (2019).
4. Kumar, D., et al.: Efficient three layer secured adaptive video steganography. Scientific Reports 14, 18301 (2024).
5. Nakamoto, S.: Bitcoin: A peer-to-peer electronic cash system. Decentralized Business Review (2008).
6. Wood, G.: Ethereum: A secure decentralised generalised transaction ledger. Ethereum project yellow paper 151(2014), 1–32 (2014).
7. Fridrich, J., Goljan, M., Du, R.: Detecting LSB steganography in color, and gray-scale images. IEEE Multimedia 8(4), 22–28 (2001).
8. Weber, A. G.: The USC-SIPI image database version 1. USC SIPI Report 315, 1 (1992).
9. Wu, X., et al.: An image steganography method based on optimized genetic algorithm. IEEE Access 8, 156–167 (2020).
10. Glover, F.: Tabu search—part I. ORSA Journal on Computing 1(3), 190–206 (1989).
11. Simmons, G.J.: The prisoners' problem and the subliminal channel. In: Advances in Cryptology: Proceedings of CRYPTO 83. Plenum Press, pp. 51–67 (1984).
12. Shannon, C.E.: Communication theory of secrecy systems. Bell System Technical Journal 28(4), 656–715 (1949).